Ensemble-Based Spatial Snow Water Equivalent Estimates Using the SWE Evaluation Tool for the Rio Grande Basin

February 22, 2026

Project Team:
Max Gersh[1], Logan Stephenson[2], Ross Palomaki[2], and Karl Rittger[2]

Contributor:
Noah Molotch[2]

[1] New Mexico Office of the State Engineer (NM OSE) Hydrology Bureau
[2] Institute of Arctic and Alpine Research (INSTAAR), University of Colorado Boulder
Work funded by: NASA Western Water Actions Office (WWAO)
Contact: Max.Gersh@ose.nm.gov

Introduction

No spaceborne sensor can directly measure snow water equivalent (SWE). Therefore, the spatial distribution of SWE must be estimated from remotely sensed observations combined with modeling approaches. Existing gridded SWE products exhibit documented accuracy limitations across both mountainous and non-mountainous regions, resulting in a wide range of SWE estimates for any given location. To address this uncertainty, the New Mexico Office of the State Engineer (NM OSE), in collaboration with the Institute of Arctic and Alpine Research (INSTAAR), developed the SWE Evaluation Tool (SWEET). SWEET integrates multiple suitable SWE data products into an ensemble framework, with the objective of producing near-real-time spatial SWE estimates for the Rio Grande Basin that are, on average, as accurate as or more accurate than any individual product alone.

Figure 1 shows the near-real-time (February 22, 2026) ensemble SWE estimates for the Rio Grande Basin. Subsequent sections of this report present detailed ensemble-based SWE maps, along with summaries by subbasin and elevation band. In addition to the ensemble results highlighted in this report, figures and tables for each individual SWE product are also provided for download to support transparency and further analysis. These data sets and visualizations are available on the GitHub repository.

For ease of navigation, hyperlinks are provided throughout this report to help readers quickly access basin-specific sections and supporting data tables.

Rio Grande Basin SWE Overview

Rio Grande Headwaters

Upper Rio Grande

Rio Grande-Elephant Butte (Middle Rio Grande)

About SWEET

Rio Grande Basin SWE Overview

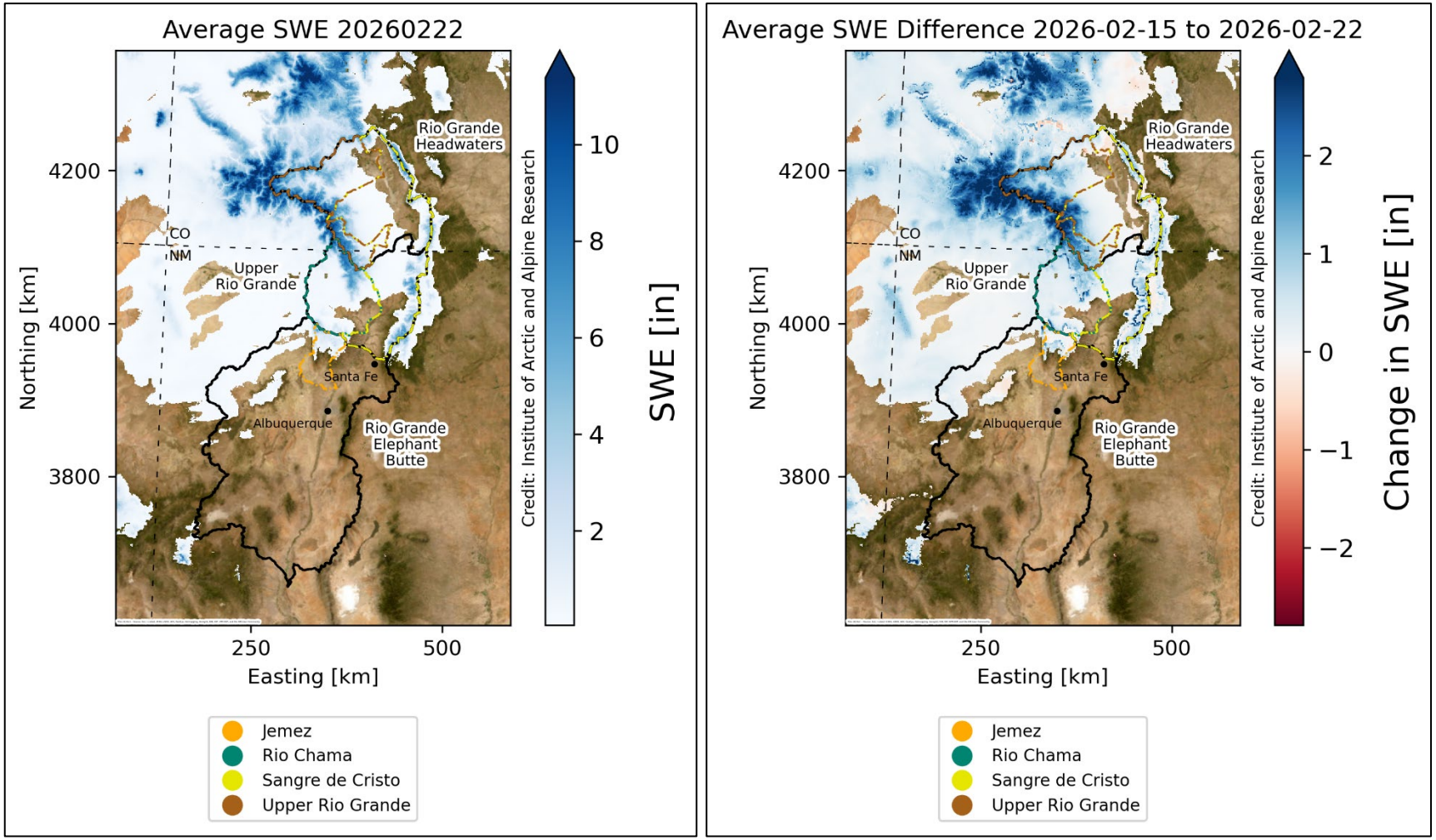

Figure 1. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Basin for February 22, 2026. *SWE amounts (in.) across the Rio Grande Basin (left); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Basin from February 15, 2026, to February 22, 2026 (right); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts.*

Table 1. Estimated ensemble SWE by HUC6 Basin for February 22, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 15, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each HUC8 basin (bold).*

HUC6 Basin	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-15 (af)	Median Vol. Change since 2026-02-15 (af)
Rio Grande Headwaters	Rio Grande Headwaters	345,416	333,334	323,402 (CU_SWE)	379,512 (SNODAS)	117,074	110,294
	Alamosa-Trinchera	131,127	138,053	102,501 (CU_SWE)	152,826 (SNODAS)	46,512	53,962
	San Luis	63,853	59,860	57,469 (CU_SWE)	74,231 (UA_SWE)	5,257	318
	Saguache	86,724	81,612	73,175 (CU_SWE)	105,384 (UA_SWE)	23,392	23,429
	Conejos	153,258	153,116	122,800 (CU_SWE)	183,858 (SNODAS)	46,223	45,290
	Total	780,377	804,118	679,347 (CU_SWE)	857,667 (SNODAS)	238,459	237,962
Upper Rio Grande	Upper Rio Grande	123,169	147,456	54,328 (CU_SWE)	167,724 (SNODAS)	21,041	36,925
	Rio Chama	195,181	182,278	149,267 (CU_SWE)	253,998 (SNODAS)	50,047	65,949
	Total	318,350	329,733	203,595 (CU_SWE)	421,722 (SNODAS)	71,088	102,874
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	2,116	1,998	1,275 (CU_SWE)	3,073 (SNODAS)	-287	996
	Jemez	23,593	23,793	15,309 (CU_SWE)	31,677 (SNODAS)	2,154	3,423
	Rio Grande-Albuquerque	95	112	0 (SNODAS)	175 (UA_SWE)	-48	-53
	Arroyo Chico	3,536	3,487	1,641 (CU_SWE)	5,482 (UA_SWE)	1,772	1,986
	North Plains	2,661	959	82 (CU_SWE)	6,943 (UA_SWE)	1,451	850
	Rio San Jose	9,613	12,084	3,696 (CU_SWE)	13,058 (UA_SWE)	4,724	9,178
	Plains of San Agustin	1,437	131	69 (CU_SWE)	4,112 (UA_SWE)	1,172	-88
	Rio Salado	290	0	0 (SNODAS, CU_SWE)	870 (UA_SWE)	278	-2
	Jornada Del Muerto	0	0	0	0	0	0
	Elephant Butte Reservoir	157	221	0 (SNODAS)	250 (UA_SWE)	-8	6
	Rio Puerco	6,050	6,979	2,119 (CU_SWE)	9,052 (UA_SWE)	276	411
	Total	49,548	58,388	24,524 (CU_SWE)	65,733 (UA_SWE)	11,485	12,848

Table 2. Estimated ensemble SWE by specified subbasin groupings for February 22, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 15, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping (bold).*

OSE Grouping	HUC8 Basin	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-15 (af)	Median Vol. Change since 2026-02-15 (af)
Upper Rio Grande	Rio Grande Headwaters	345,416	333,334	323,402 (CU_SWE)	379,512 (SNODAS)	117,074	110,294
	Saguache	86,724	81,612	73,175 (CU_SWE)	105,384 (UA_SWE)	23,392	23,429
	Conejos	153,258	153,116	122,800 (CU_SWE)	183,858 (SNODAS)	46,223	45,290
	Total	585,398	591,834	519,377 (CU_SWE)	644,982 (SNODAS)	186,690	178,854
Sangre de Cristo	San Luis	63,853	59,860	57,469 (CU_SWE)	74,231 (UA_SWE)	5,257	318
	Alamosa-Trinchera	131,127	138,053	102,501 (CU_SWE)	152,826 (SNODAS)	46,512	53,962
	Upper Rio Grande	123,169	147,456	54,328 (CU_SWE)	167,724 (SNODAS)	21,041	36,925
	Total	318,149	359,739	214,298 (CU_SWE)	380,410 (SNODAS)	72,810	85,576
Rio Chama	**Total**	195,181	182,278	149,267 (CU_SWE)	253,998 (SNODAS)	50,047	65,949
Jemez	**Total**	23,593	23,793	15,309 (CU_SWE)	31,677 (SNODAS)	2,154	3,423

Upper Rio Grande Snowpack Update Application

The OSE/ISC Upper Rio Grande Snowpack Update Application (Application) provides water managers and users with detailed SNOTEL-based SWE data organized by the same subbasin groupings shown in Table 2. The Application is updated regularly from December through May and compiles key SWE information from SNOTEL sites within each subbasin. In addition to the Upper Rio Grande groupings (Upper Rio Grande, Sangre de Cristo, Rio Chama, and Jemez), the Application also includes the San Juan River Basin, with data summarized for the Upper San Juan, Animas, and Piedra subbasins.

The Application can be viewed here[1].

[1] Please note that access to the Application may be restricted to employees of the NM OSE/ISC at this time. We are currently working on public access.

Rio Grande Headwaters

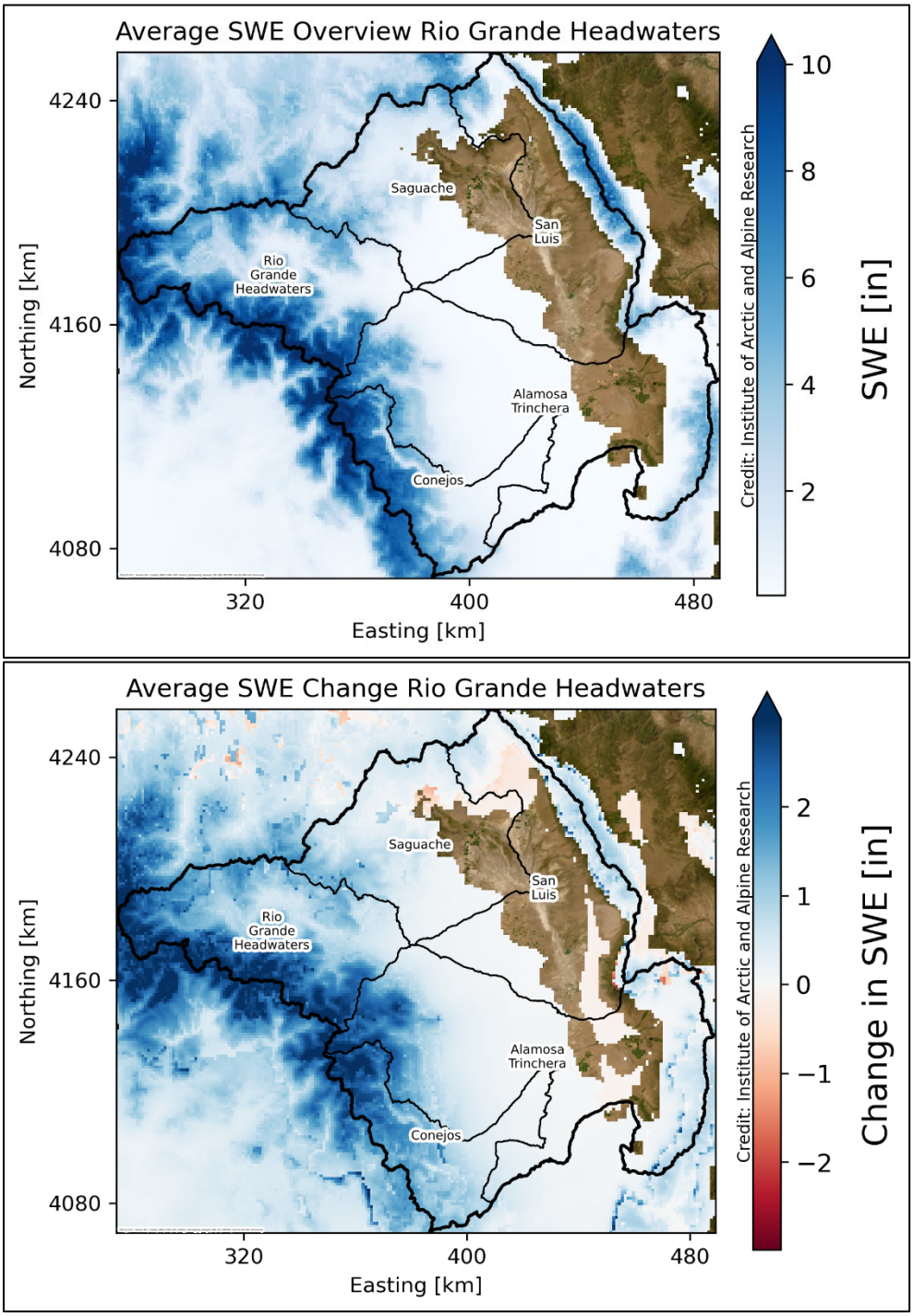

Figure 2. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande Headwaters HUC6 Basin for February 22, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande Headwaters Basin, showing spatial patterns of SWE across subbasins including Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande Headwaters from February 15, 2026, to February 22, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

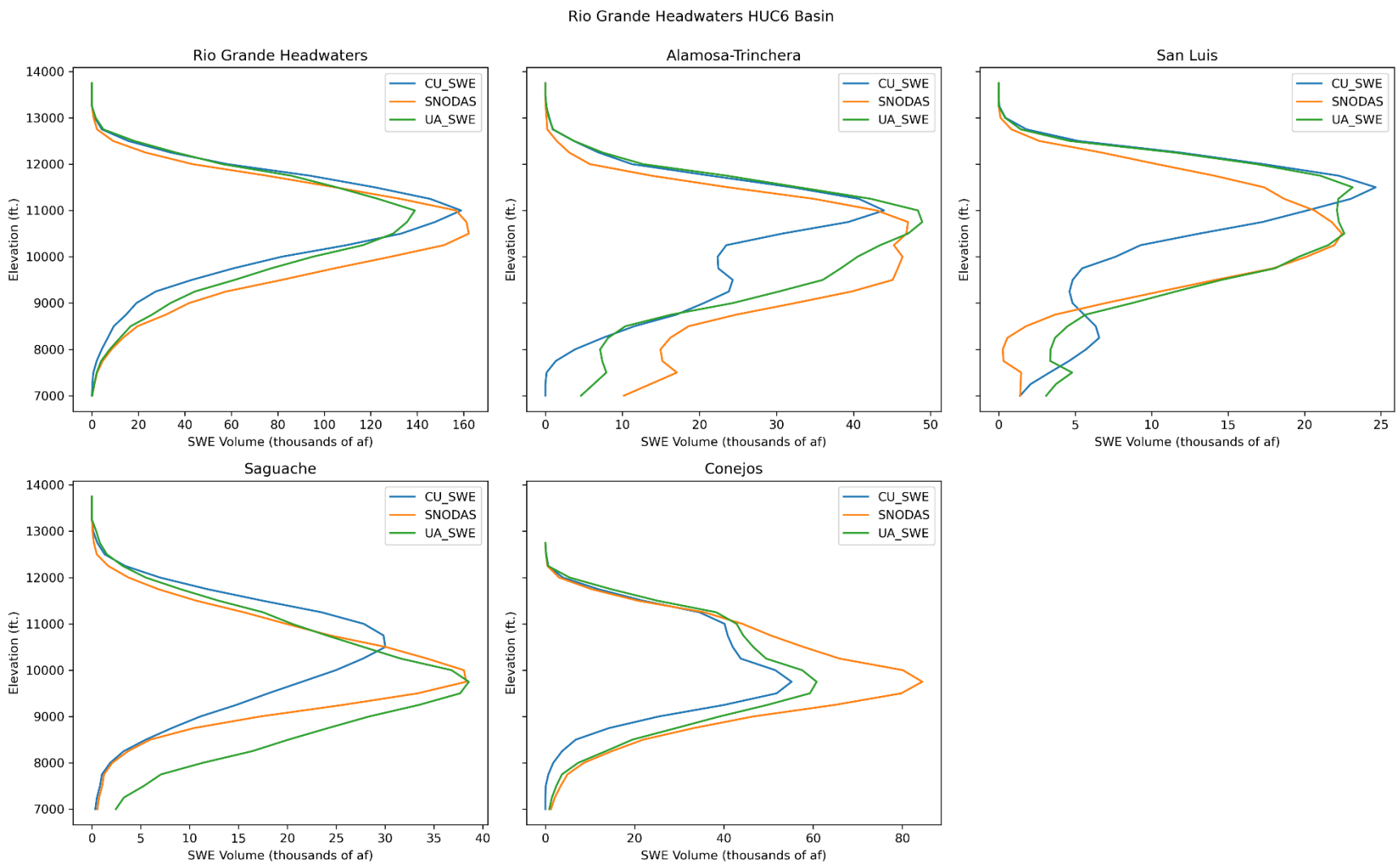

Figure 3. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for February 22, 2026. *Elevation-distributed SWE volume for the Rio Grande Headwaters HUC6 Basin and associated subbasins (Rio Grande Headwaters, Alamosa–Trinchera, San Luis, Saguache, and Conejos), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 3. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande Headwaters HUC6 Basin for February 22, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 15, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-15 (af)	Median Vol. Change since 2026-02-15 (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	83	115	0 (CU_SWE)	135 (SNODAS)	81	115
		8000-9000'	6,757	7,649	4,201 (CU_SWE)	8,419 (SNODAS)	5,314	6,233
		9000-10000'	31,587	33,799	19,144 (CU_SWE)	41,819 (SNODAS)	16,017	14,230
		10000-11000'	101,737	95,048	81,634 (CU_SWE)	128,530 (SNODAS)	36,416	33,529
		11000-12000'	151,516	156,668	138,977 (UA_SWE)	158,904 (CU_SWE)	45,270	52,174
		12000-13000'	52,555	56,128	43,251 (SNODAS)	58,286 (CU_SWE)	13,720	14,925
		13000-14000'	1,180	1,232	691 (SNODAS)	1,618 (UA_SWE)	257	160
	Alamosa-Trinchera	7000-8000'	4,935	4,620	0 (CU_SWE)	10,186 (SNODAS)	2,892	4,045
		8000-9000'	8,634	7,110	3,846 (CU_SWE)	14,947 (SNODAS)	6,107	5,462
		9000-10000'	25,773	24,333	20,574 (CU_SWE)	32,412 (SNODAS)	10,481	10,127
		10000-11000'	36,411	40,510	22,362 (CU_SWE)	46,360 (SNODAS)	12,692	12,709
		11000-12000'	45,105	43,959	42,996 (SNODAS)	48,360 (UA_SWE)	12,465	9,238
		12000-13000'	9,895	11,241	5,778 (SNODAS)	12,665 (UA_SWE)	1,847	1,478
		13000-14000'	374	455	147 (SNODAS)	520 (CU_SWE)	26	74
	San Luis	7000-8000'	1,963	1,400	1,391 (CU_SWE)	3,100 (UA_SWE)	-2,089	-93
		8000-9000'	3,117	3,389	258 (SNODAS)	5,704 (CU_SWE)	-91	664
		9000-10000'	6,846	6,994	4,823 (CU_SWE)	8,722 (UA_SWE)	907	-343
		10000-11000'	15,795	19,622	7,641 (CU_SWE)	20,122 (SNODAS)	2,088	2,881
		11000-12000'	20,979	20,640	20,177 (CU_SWE)	22,120 (UA_SWE)	2,886	2,101
		12000-13000'	14,830	16,864	10,333 (SNODAS)	17,292 (CU_SWE)	1,536	3,065
		13000-14000'	323	415	113 (SNODAS)	441 (CU_SWE)	20	76
	Saguache	7000-8000'	1,117	560	342 (CU_SWE)	2,449 (UA_SWE)	207	188
		8000-9000'	5,099	2,080	1,868 (CU_SWE)	11,350 (UA_SWE)	1,333	1,121
		9000-10000'	18,888	17,214	11,090 (CU_SWE)	28,362 (UA_SWE)	5,932	5,639
		10000-11000'	33,281	36,807	24,963 (CU_SWE)	38,071 (SNODAS)	9,178	8,252
		11000-12000'	22,728	20,474	19,884 (SNODAS)	27,825 (CU_SWE)	5,519	5,153
		12000-13000'	5,403	5,503	3,746 (SNODAS)	6,959 (CU_SWE)	1,188	1,426
		13000-14000'	208	128	56 (SNODAS)	439 (UA_SWE)	34	11
	Conejos	7000-8000'	719	891	0 (CU_SWE)	1,266 (SNODAS)	624	891

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-15 (af)	Median Vol. Change since 2026-02-15 (af)
Rio Grande Headwaters	Conejos	8000-9000'	5,925	7,360	1,770 (CU_SWE)	8,645 (SNODAS)	3,975	4,540
		9000-10000'	36,989	39,021	25,397 (CU_SWE)	46,548 (SNODAS)	11,743	12,301
		10000-11000'	63,078	57,557	51,537 (CU_SWE)	80,140 (SNODAS)	16,897	14,884
		11000-12000'	42,404	42,862	40,174 (CU_SWE)	44,175 (SNODAS)	11,848	11,761
		12000-13000'	4,144	3,923	3,083 (SNODAS)	5,425 (UA_SWE)	1,135	690

Upper Rio Grande

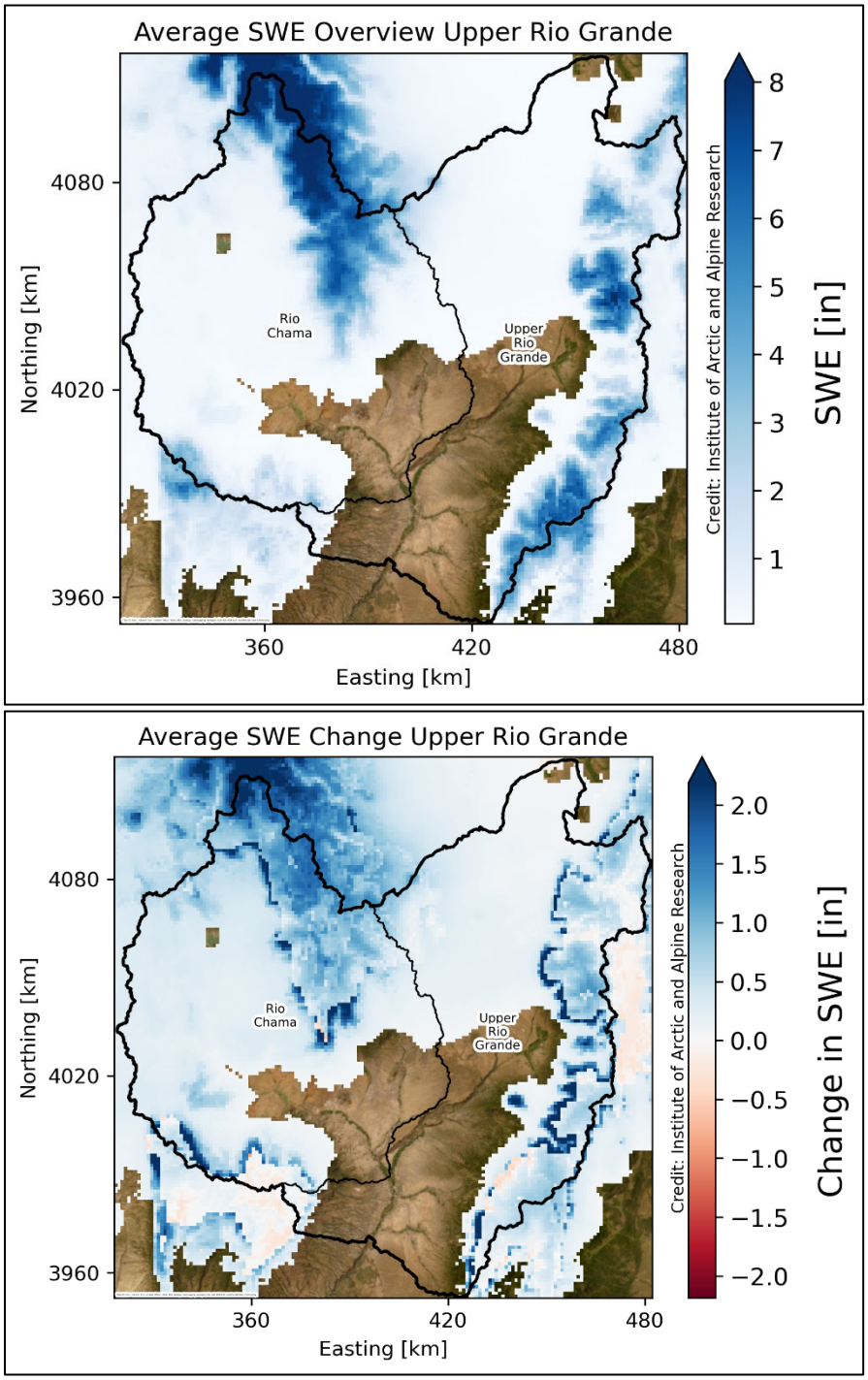

Figure 4. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Upper Rio Grande HUC6 Basin for February 22, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Upper Rio Grande Basin, showing spatial patterns of SWE across subbasins including Rio Chama and Upper Rio Grande (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Upper Rio Grande Basin from February 15, 2026, to February 22, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

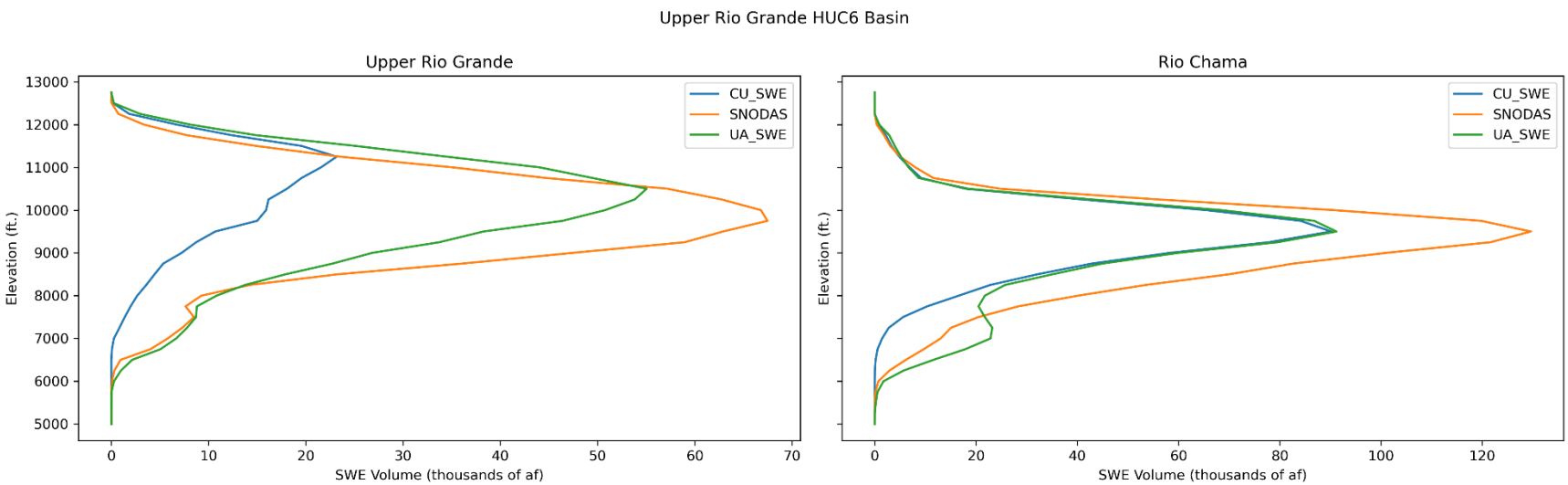

Figure 5. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Upper Rio Grande HUC6 Basin for February 22, 2026. *Elevation-distributed SWE volume for the Upper Rio Grande HUC6 Basin and associated subbasins (Rio Chama and Upper Rio Grande), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 4. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Upper Rio Grande HUC6 Basin for February 22, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 15, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-15 (af)	Median Vol. Change since 2026-02-15 (af)
Upper Rio Grande	Upper Rio Grande	5000-6000'	0	0	0	0	0	0
		6000-7000'	115	70	0 (CU_SWE)	277 (UA_SWE)	115	70
		7000-8000'	4,243	5,794	266 (CU_SWE)	6,669 (UA_SWE)	3,532	4,927
		8000-9000'	7,578	9,253	2,664 (CU_SWE)	10,817 (UA_SWE)	2,010	3,934
		9000-10000'	27,148	26,808	7,222 (CU_SWE)	47,413 (SNODAS)	3,540	6,485
		10000-11000'	44,487	50,765	15,911 (CU_SWE)	66,786 (SNODAS)	6,768	9,941
		11000-12000'	33,551	35,072	21,560 (CU_SWE)	44,023 (UA_SWE)	4,397	3,470
		12000-13000'	6,046	6,705	3,336 (SNODAS)	8,097 (UA_SWE)	679	125
	Rio Chama	5000-6000'	0	0	0	0	0	0
		6000-7000'	820	725	9 (CU_SWE)	1,727 (UA_SWE)	812	725
		7000-8000'	12,439	13,009	1,454 (CU_SWE)	22,854 (UA_SWE)	9,000	12,676
		8000-9000'	26,160	21,745	16,590 (CU_SWE)	40,144 (SNODAS)	8,580	11,786
		9000-10000'	73,049	59,910	58,249 (CU_SWE)	100,988 (SNODAS)	14,714	13,170
		10000-11000'	74,902	68,469	65,512 (CU_SWE)	90,726 (SNODAS)	15,111	13,836
		11000-12000'	7,203	6,914	6,681 (UA_SWE)	8,015 (SNODAS)	1,689	1,700
		12000-13000'	607	539	392 (SNODAS)	891 (UA_SWE)	141	113

Rio Grande-Elephant Butte (Middle Rio Grande)

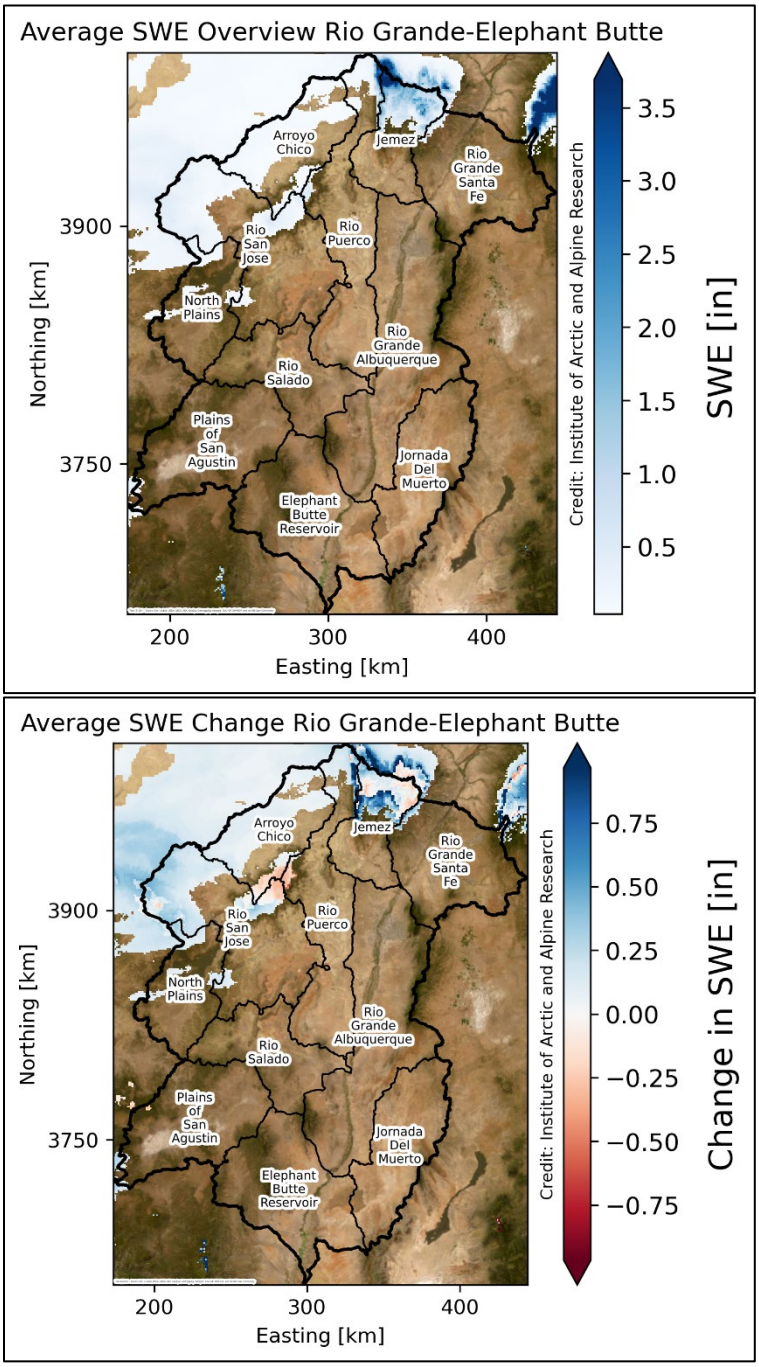

Figure 6. Estimated ensemble SWE (in.) and change in estimated ensemble SWE (in.) across the Rio Grande-Elephant Butte HUC6 Basin for February 22, 2026, derived from the SWE Evaluation Tool (SWEET). *Ensemble SWE amount across the Rio Grande-Elephant Butte Basin, showing spatial patterns of SWE across subbasins including Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto (top); dark blue pixels denote greater SWE amounts. Change in SWE amounts (in.) across the Rio Grande-Elephant Butte Basin from February 15, 2026, to February 22, 2026 (bottom); blue pixels denote an increase in SWE amounts whereas red pixels denote a decrease in SWE amounts. Basin and subbasin boundaries are shown in black.*

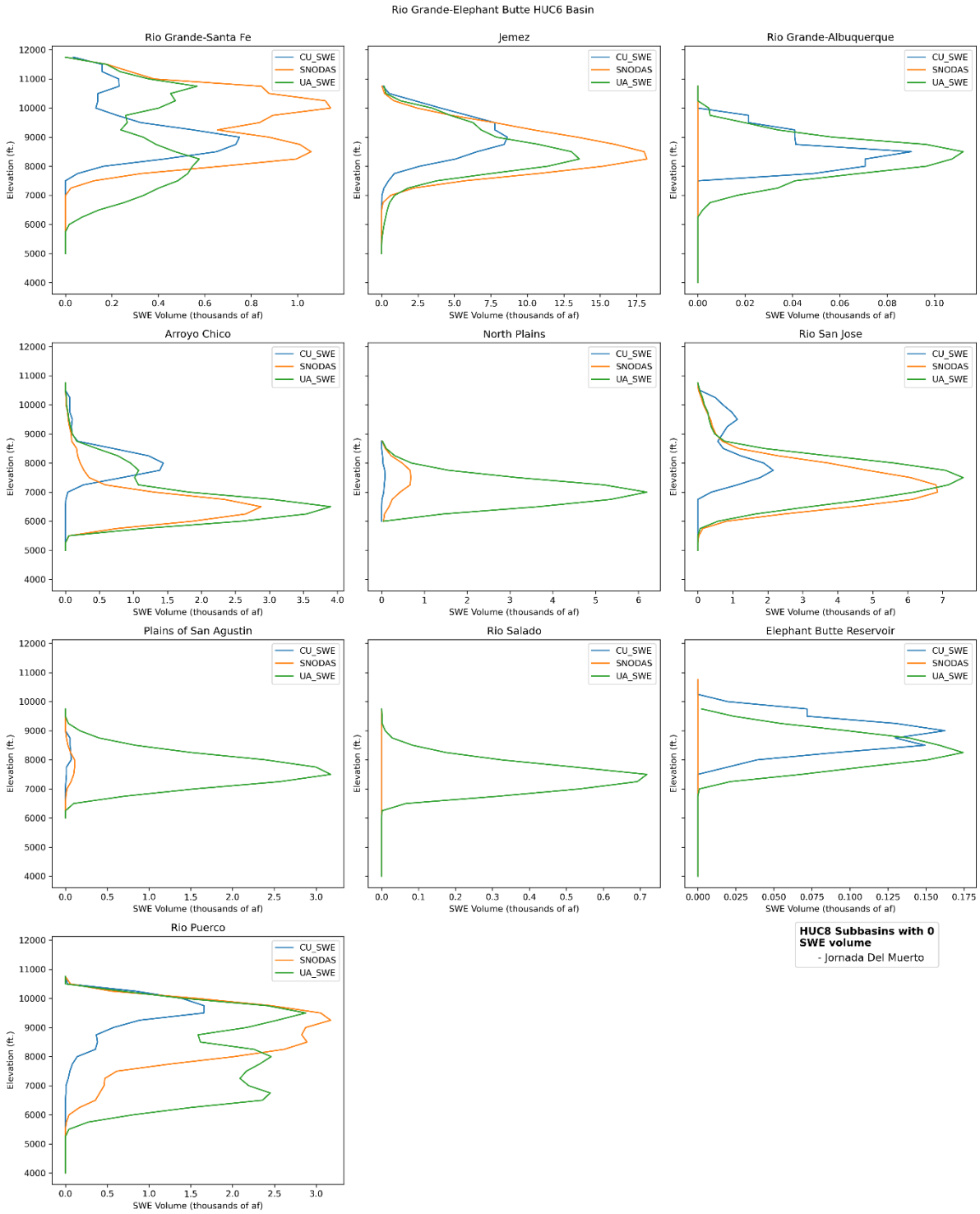

Figure 7. Estimated SWE volume (thousands of af) by subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for February 22, 2026. *Elevation-distributed SWE volume for the Rio Grande-Elephant Butte HUC6 Basin and associated subbasins (Rio Grande-Santa Fe, Jemez, Arroyo Chico, Rio San Jose, Rio Puerco, Rio Grande-Albuquerque, Rio Salado, North Plains, Plains of San Agustin, Elephant Butte Reservoir, and Jornada Del Muerto), comparing three spatial SWE products (CU_SWE, SNODAS, and UA_SWE). Curves show SWE volume summarized by elevation band, illustrating differences in both magnitude and vertical distribution among products across each subbasin.*

Table 5. Estimated ensemble SWE by HUC8 subbasin and elevation band across the Rio Grande-Elephant Butte HUC6 Basin for February 22, 2026. *Shown are ensemble mean and median volume (af), ensemble minimum volume (af), ensemble maximum volume (af), mean and median volume change (af) since February 15, 2026. The SWE product in parenthesis in the minimum and maximum columns represents which product is reporting the value.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-15 (af)	Median Vol. Change since 2026-02-15 (af)
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0	0	0	0	0	0
		6000-7000'	6	0	0 (SNODAS, CU_SWE)	17 (UA_SWE)	6	0
		7000-8000'	113	1	0 (CU_SWE)	337 (UA_SWE)	-11	1
		8000-9000'	461	547	164 (CU_SWE)	671 (SNODAS)	-381	451
		9000-10000'	654	750	335 (UA_SWE)	876 (SNODAS)	41	308
		10000-11000'	558	402	131 (CU_SWE)	1,142 (SNODAS)	40	10
		11000-12000'	324	360	230 (CU_SWE)	383 (SNODAS)	19	4
	Jemez	5000-6000'	0	0	0	0	0	0
		6000-7000'	58	0	0 (SNODAS, CU_SWE)	173 (UA_SWE)	57	0
		7000-8000'	512	623	21 (CU_SWE)	891 (UA_SWE)	-65	572
		8000-9000'	9,724	11,389	2,606 (CU_SWE)	15,177 (SNODAS)	251	860
		9000-10000'	9,979	8,629	7,873 (UA_SWE)	13,437 (SNODAS)	1,370	1,697
		10000-11000'	3,320	3,467	2,440 (SNODAS)	4,053 (CU_SWE)	541	610
	Rio Grande-Albuquerque	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	6	0	0 (SNODAS, CU_SWE)	17 (UA_SWE)	6	0
		8000-9000'	56	71	0 (SNODAS)	97 (UA_SWE)	-5	-17
		9000-10000'	33	41	0 (SNODAS)	57 (UA_SWE)	-43	-36
		10000-11000'	2	0	0 (SNODAS, CU_SWE)	5 (UA_SWE)	-5	0
	Arroyo Chico	5000-6000'	0	0	0	0	0	0
		6000-7000'	1,485	1,869	0 (CU_SWE)	2,586 (UA_SWE)	1,449	1,866
		7000-8000'	1,053	1,305	36 (CU_SWE)	1,816 (UA_SWE)	660	826
		8000-9000'	872	958	218 (SNODAS)	1,440 (CU_SWE)	-371	-182
		9000-10000'	96	99	86 (SNODAS)	104 (UA_SWE)	23	18
		10000-11000'	30	17	9 (SNODAS)	65 (CU_SWE)	12	6
	North Plains	6000-7000'	31	46	0 (CU_SWE)	46 (SNODAS)	31	46
		7000-8000'	2,227	432	51 (CU_SWE)	6,199 (UA_SWE)	1,131	427
		8000-9000'	403	481	31 (CU_SWE)	698 (UA_SWE)	289	377

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-15 (af)	Median Vol. Change since 2026-02-15 (af)
Rio Grande-Elephant Butte	Rio San Jose	5000-6000'	0	0	0	0	0	0
		6000-7000'	465	575	0 (CU_SWE)	819 (SNODAS)	313	575
		7000-8000'	4,478	6,194	385 (CU_SWE)	6,855 (SNODAS)	2,766	6,190
		8000-9000'	3,746	3,736	1,884 (CU_SWE)	5,618 (UA_SWE)	1,295	1,507
		9000-10000'	568	513	484 (UA_SWE)	708 (CU_SWE)	130	105
		10000-11000'	356	188	161 (SNODAS)	719 (CU_SWE)	220	48
	Plains of San Agustin	6000-7000'	0	0	0	0	0	0
		7000-8000'	521	19	0 (CU_SWE)	1,545 (UA_SWE)	518	17
		8000-9000'	858	110	69 (CU_SWE)	2,394 (UA_SWE)	658	-65
		9000-10000'	58	2	0 (CU_SWE)	173 (UA_SWE)	-4	-35
	Rio Salado	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	179	0	0 (SNODAS, CU_SWE)	536 (UA_SWE)	179	0
		8000-9000'	108	0	0 (SNODAS, CU_SWE)	324 (UA_SWE)	99	-1
		9000-10000'	3	0	0 (SNODAS, CU_SWE)	10 (UA_SWE)	0	-1
	Jornada Del Muerto	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0	0	0	0
		8000-9000'	0	0	0	0	0	0
	Elephant Butte Reservoir	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	0	0	0	0	0	0
		7000-8000'	0	0	0 (SNODAS, CU_SWE)	1 (UA_SWE)	0	0
		8000-9000'	63	39	0 (SNODAS)	151 (UA_SWE)	19	0
		9000-10000'	87	98	0 (SNODAS)	162 (CU_SWE)	-27	-67
		10000-11000'	10	10	0 (SNODAS)	20 (CU_SWE)	-1	-1
	Rio Puerco	4000-5000'	0	0	0	0	0	0
		5000-6000'	0	0	0	0	0	0
		6000-7000'	284	44	0 (CU_SWE)	809 (UA_SWE)	265	44
		7000-8000'	887	462	7 (CU_SWE)	2,193 (UA_SWE)	44	119

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Ensemble Mean Vol. (af)	Ensemble Median Vol. (af)	Ensemble Min. Vol. (af)	Ensemble Max. Vol. (af)	Mean Vol. Change since 2026-02-15 (af)	Median Vol. Change since 2026-02-15 (af)
Rio Grande-Elephant Butte	Rio Puerco	8000-9000'	1,540	2,018	142 (CU_SWE)	2,461 (UA_SWE)	-450	-725
		9000-10000'	1,872	2,169	575 (CU_SWE)	2,873 (SNODAS)	208	253
		10000-11000'	1,466	1,420	1,395 (CU_SWE)	1,582 (SNODAS)	208	294

About SWEET

The following information provides a brief overview of SWEET, and the SWE products currently included in the ensemble framework. For more detailed descriptions, citations, and additional information, please refer to the GitHub repository.

SWE products currently used include:

1. **SNODAS SWE**
    a. The SNOw Data Assimilation System (SNODAS) is a physically based snow model developed by NOAA's National Operational Hydrologic Remote Sensing Center and operational since 2004. SNODAS produces daily 1-km gridded estimates of SWE and related snow variables by combining modeled weather inputs with satellite, airborne, and in situ snow observations.
2. **UA/SWANN SWE**
    a. The University of Arizona/Snow Water Artificial Neural Network Modeling System (UA/SWANN) is a research SWE product that uses snow models, artificial neural networks, and assimilated in situ measurements to generate daily gridded SWE and snow depth. The dataset provides 4-km and downscaled 800-m SWE across the conterminous United States, incorporating SNOTEL and COOP observations with PRISM precipitation and temperature fields.
3. **CU SWE**
    a. The University of Colorado (CU) SWE-Fusion product is an operational dataset produced by INSTAAR that provides near–real-time SWE estimates at 500-m resolution. The approach fuses SNOTEL observations with historical SWE reanalyses and terrain variables using a generalized linear model, and scales results using satellite-derived fractional snow-covered area from Snow Today.

The following notes describe key processing methods, assumptions, and visualization conventions used in generating SWE Evaluation Tool figures, tables, and elevation plots.

Figures:

- For ensemble figures, CU SWE and UA SWE are reprojected to the SNODAS resolution (1 sq km) and averages are only taken at pixels where at least 2 of the 3 products have SWE values greater than 0 inches.
- All overview and subbasin plots set pixels with values of 0 inches to NA so that the basemap is visible.
- For individual product figures, each product is displayed at its native resolution.
- Elevation bands of 250 feet are used in the elevation figures, and SWE volume is calculated for each product's native resolution in each band.
- Difference plots display pixels where snow was present in either the current or previous report.
- Figures are available in PNG format on the GitHub repository.

Tables:

- For all tables, mean, median, and volume are calculated at each product's native resolution - zeros are not removed from mean and median calculations.
- For ensemble tables, mean and median volume are calculated using the total volumes from the 3 products - not from an average aggregate raster.
- For min. and max., if 2 products have 0 volume, both products are listed in that cell. If all 3 products have 0 volume, no product is listed for min. or max. in that row.
- Change in SWE will only be calculated starting with the second report of the water year.
- Ensemble calculations: For HUC8 subbasins, the ensemble (mean and median) SWE and volume change (mean and median) are calculated individually for each subbasin. For the HUC6 "Total" row (tables 1 and 2), the ensemble is recalculated using the full HUC6 basin extent rather than summing HUC8. While the ensemble mean is additive and therefore consistent between HUC8 sums and the HUC6 total, the median reflects the middle value among the three products for the entire basin. As a result, median values will not sum to the HUC6 total, since they represent the difference between basin-wide medians rather than the sum of subbasin medians.
- Tables are available in CSV and PDF format on the GitHub repository.
- Elevation tables presented in the report (Tables 3, 4, and 5) are combined into one table (Table 3) on the GitHub repository.

Acknowledgments

The work completed and presented in this report is funded by NASA Western Water Actions Office (WWAO), subcontract no. 1712706, as part of the *Rio Grande River Basin Needs Assessment Report*. The SWE Evaluation Tool is a collaborative effort from members at INSTAAR, University of Colorado Boulder, and the New Mexico Office of the State Engineer.

We would also like to extend our appreciation to Noah Molotch and the Mountain Hydrology Group at CU Boulder, INSTAAR, for granting our team access to the CU SWE operational SWE dataset.